

Leopold Mareis

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About

I aim to develop reliable and explainable statistical methods for estimating causal effects. This requires valid confidence statements throughout the study, from data collection to final decision. My work addresses these questions with problems encountered in practice: experimental design with measurement constraints, missing data, and inference that connects to actual decisions. My perspective comes from direct collaborations in medicine, social science, and industry.

Education

- **Doctoral Candidate**, Mathematical Statistics, TU Munich, Germany (12/2021 - 12/2026)
Thesis: *Asymptotic Inference in Linear Causal Models*
Supervisor: Prof. Dr. Mathias Drton
- **Master of Mathematics in Operations Research**, TU Munich, Germany (2018–2021)
Thesis: *Vine copula based quantile regression including discrete components*
Supervisor: Prof. Dr. Claudia Czado
- **Semester Abroad**, TU Delft, Netherlands (2019 - 2020)
- **Bachelor of Mathematics**, TU Munich, Germany (2015–2018)
Thesis: *Inexact successive quadratic approximation methods*

Employment

- **Scientific Researcher**, Technical University of Munich, DE (since 12/2024)
 - Funded through the *Mathematical Research Data Initiative (MaRDI)* by NFDI.
 - Developed and maintained the Graphical Modeling and Causal Inference community and website.
 - Curated topical datasets and authored instructional notebooks on computational causality.
- **Scientific Researcher**, Fraunhofer Institute for Cognitive Systems IKS, DE (10/2021 - 11/2024)
 - Department: Reasoned AI Decisions
 - Developed counterfactual RL decision support for the ADVITOS multi-organ support system.
 - Conducted causal analyses for the BIOTRONIK BIOSOLVE-II clinical trial.
 - Built causal inference libraries (R and Python) for industry applications.
- **Teaching Assistant**, TU Munich, DE
 - Computational Statistics (Summer 2026)
 - Graphical Models in Statistics (Summer 2025)
 - Seminar: Machine Learning for Sequential Decision Making (Winter 2023)
 - Statistics for Business Administration (Summer 2020)
 - Analysis 1 for Physicists (Winter 2016)
- **Secretary**, TU Munich IAS, DE (Summer 2016)
- **Internships**: MunichRe (2020), Agrilution (2018)

Publications

- **Leopold Mareis**, Nils Sturma, Mathias Drton. *Semiparametric Inference for Half-Trek Estimators in Linear Structural Equation Models*. Accepted at *Probabilistic Graphical Models (PGM) 2026*.
- Ilse Hagerer, **Leopold Mareis**, Daniela Schkoda. *Causes and Effectiveness of Gender Equality Instruments in Organizations within Germany*. Under review.
- **Leopold Mareis**, Mathias Drton. *Cost-Aware Optimized Front-Door Experimental Design*. Accepted at *Causal Learning and Reasoning (CLear) 2026*.
- **Leopold Mareis**, Stephan Haug, Mathias Drton. *MaRDI's Zenodo Community for Graphical Modeling and Causal Inference*. 2nd Conference on Research Data Management. 2025
- Jan-Angelus Meyer, Jennifer Pierick, **Leopold Mareis**, Dorian Röhring, Nathalie Opilo, Georg Garnweitner, Rainer Krull. *Low-gradient magnetic separation in Biotechnology: Parameter identification and model-guided scale up*. Separation and Purification Technology. 2025
- **Leopold Mareis**. *Optimizing Experimental Design for Causal Effect Estimation with Partial Measurements*. 2nd Pacific Causal Inference Conference proceedings, 2024.
- Narges Ahmidi, **Leopold Mareis**. *Development and certification of clinical AI software*. 2024

Presentations

Oral:

- PGM 2026: *Semiparametric Inference for Half-Trek Estimators in Linear Structural Equation Models* (09/2026)
- JuliaCon 2026: *Graphical Modeling with Symbolic Algebra in OSCAR.jl* (08/2026)
- IMS Annual Meeting: *Cost-Aware Optimized Front-Door Experimental Design* (07/2026)
- Statistical Computing Conference Ulm 2025: *Ontology for R language packages* (07/2025)
- bitkom - AI & Data Summit: *Counterfactual decision support for the control of a clinical, multi-organ failure ICU machine* (09/2024)
- Doktorand:innentreffen der Stochastik: *Graph Projections for Causal DAGs* (08/2023)
- KI-Medtech Congress: *Estimating Causal Effects of Stent Operations* (07/2023)
- ETH-UCPH-TUM Workshop: *Nonparametric Causal Effect Estimation of Guideline Interventions* (10/2022)
- Doktorand:innentreffen der Stochastik: *Nonparametric Causal Effect Estimation* (08/2022)

Poster:

- Conference on Causal Learning and Reasoning CLear 2026: *Cost-Aware Optimized Front-Door Experimental Design* (04/2026)
- 2nd Conference on Research Data Infrastructure: *MaRDI's Zenodo Community for Graphical Modeling and Causal Inference* (09/2025)
- 11th World Congress in Probability and Statistics: *Adaptive Experimental Design for Optimal Causal Effect Estimation* (08/2024)
- Oxford StatML Summer School: *Nonparametric Causal Effect Estimation of Guideline Interventions* (08/2023)
- AI-Bay Conference: *Nonparametric Causal Effect Estimation of Guideline Interventions* (02/2023)

Students

Master's Theses:

- Žiga Gartner: *Reanalyzing the DREAM 5 data using techniques from graphical modeling* (Summer 2026)
- Darya Shtromberg: *Graphical Dependence Modeling in Stock Market Analysis* (Winter 2025)

- Adrien Castella: *Active Learning for ML-training for Autonomous Vehicles* (Summer 2024)
- Qianyun Li: *Graph-Simulation Based Analysis of Causal Inference Methods on Instrumental Variables* (Summer 2023)

University Seminar:

- Ronald Skorbogad: *Estimating Counterfactual Treatment Outcomes Through Adversarially Balanced Representations* (Winter 2023)
- Dogu Tomac: *An Overview of Continuous-Time Models and Their Applications* (Winter 2023)

Working Students:

- Sarah Lumpp (10/21 – 07/23)
- Daniel Sturm (10/21 – 11/22)

Computer and Language Skills

Programming: R (Advanced), Python (Intermediate), Julia, C++ and SageMath (Beginner)
Software: PARCS - *Partially Randomized Causal Simulator* (Python),
<https://fraunhoferiks.github.io/parcs/>, with accompanying arXiv preprint
Website: *Graphical Modeling and Causal Inference*. <https://st-mardi.quarto.pub/gmci/>
Languages: German (Mother tongue), English (C1), French (B1)

Personal

My big passion is classical music and in particular the choir. I am a state-certified choir conductor and am the elected youth referent in the Bavarian choir association (BSB) since October 2024.